Brightlights Data Analytics

SQL Exercise: Aggregate Functions and Grouping

① SELECT COUNT(id) AS total_employees
FROM employees;

total_employees
10

② SUM() Function
Write a SQL query to find the total salary paid to all employees in IT department

```
SELECT
        SUM(salary) AS total_payroll
FROM employees
WHERE department='IT';
```

total_payroll
220000

③ SELECT
AVG(Salary) AS Average_Salary
FROM employees
WHERE department='HR';

Average_Salary
49500

④ SELECT MAX(Salary) AS highest_salary,
MIN(Salary) AS lowest_Salary
FROM employees;

Highest_salary	lowest Salary
62000	48000

⑤ SELECT ~~first_name~~,
department,
SUM(Salary) AS total_department_Salary
FROM employees
GROUP BY department;

~~First_name~~	Department	total_department Salary
	IT	220000
	HR	99000
	Finance	119000
	Marketing	105000

6. SELECT city,
SUM(id) AS number_of_employees
FROM employees
GROUP B city;

City	Number_of_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. SELECT department,
AVG(salary) AS Average_salary
FROM employees
~~WHERE~~
ORDER By Average_salary;

department	Average_salary
IT	55000
HR	49500
Finance	59500
Marketing	52500

8. SELECT department
SUM(salary) AS total_department_salary
FROM employees
GROUP By department
~~Hav~~ HAVING SUM(salary) > 100000;

department	total_department_salary
IT	220000
Finance	119000
Marketing	105000

④ SELECT city
COUNT(id) AS employee_count
FROM employees

GROUP BY city
HAVING COUNT(id) > 1
ORDER BY employee_count DESC;

city	employee_count
~~San Francisco~~	
Chicago	3
New York	2
Los Angeles	2
San Jose	2